Embedded and Cyber Physical System Components

The I^{2}C Bus

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Lecture References:

- Book:
  - *Embedded System Design*, **Author:** P. Marwedel
  - *Embedded System Design: An Introduction to Processes, Tools and Techniques*, **Author:** Arnold Berger, Arnold S. Berger
- Lecture Materials:
  - *Microprocessor Engineering,* Sheffield Halam University.

The I^2C Bus

→ Inter IC bus
→ for small scale embedded devices

- What is the I^2C Bus and what is it used for?
- Bus characteristics
- I^2C Bus Protocol
- Data Format
- Typical I^2C devices
- Example device

What is I²C

- **The name stands for "Inter - Integrated Circuit Bus"**
- **A Small Area Network connecting ICs and other electronic components in an embedded systems**
- **Originally intended for operation on one single board / PCB having features like:**
  - Synchronous Serial Signal
  - Two wires carry information between a number of devices
  - One wire use for the data
  - One wire used for the clock → to maintain same clo
- **Today, a variety of devices are available with I²C Interfaces**
  - Microcontroller, EEPROM, Real-Timer, interface chips, LCD driver, A/D converter

What is I^{2}C used for?

- **Data transfer between ICs and systems at relatively low rates**
  - "Classic" I^{2}C is rated to 100K bits/second
  - "Fast Mode" devices support up to 400K bits/second
  - "High Speed Mode" is defined for operation up to 3.4 Mbits/second

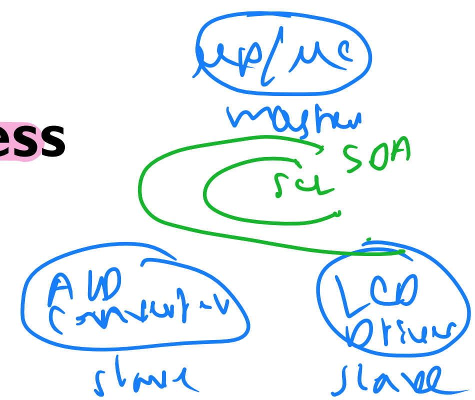
- **Reduces Board Space and Cost By:**
  - Allowing use of ICs with fewer pins and smaller packages , smaller board as well
  - Greatly reducing interconnect complexity
  - Allowing digitally controlled components to be located close to their point of use

→ EMI is negligible

I²C Bus Characteristics

- **Includes electrical and timing specifications, and an associated bus protocol**
- **Two wire serial data & control bus implemented with the serial data (SDA) and clock (SCL) lines**
  - For reliable operation, a third line is required: Common ground
- **Unique start and stop condition**

→ $2^7 = 128$ bits



- **Slave selection protocol uses a 7-Bit slave address**
  - The bus specification allows an extension to 10 bits
- **Bi-directional data transfer**
- **Acknowledgement after each transferred byte**
- **No fixed length of transfer**

CAN had 8 bytes

I^2C Bus Characteristics (cont'd)

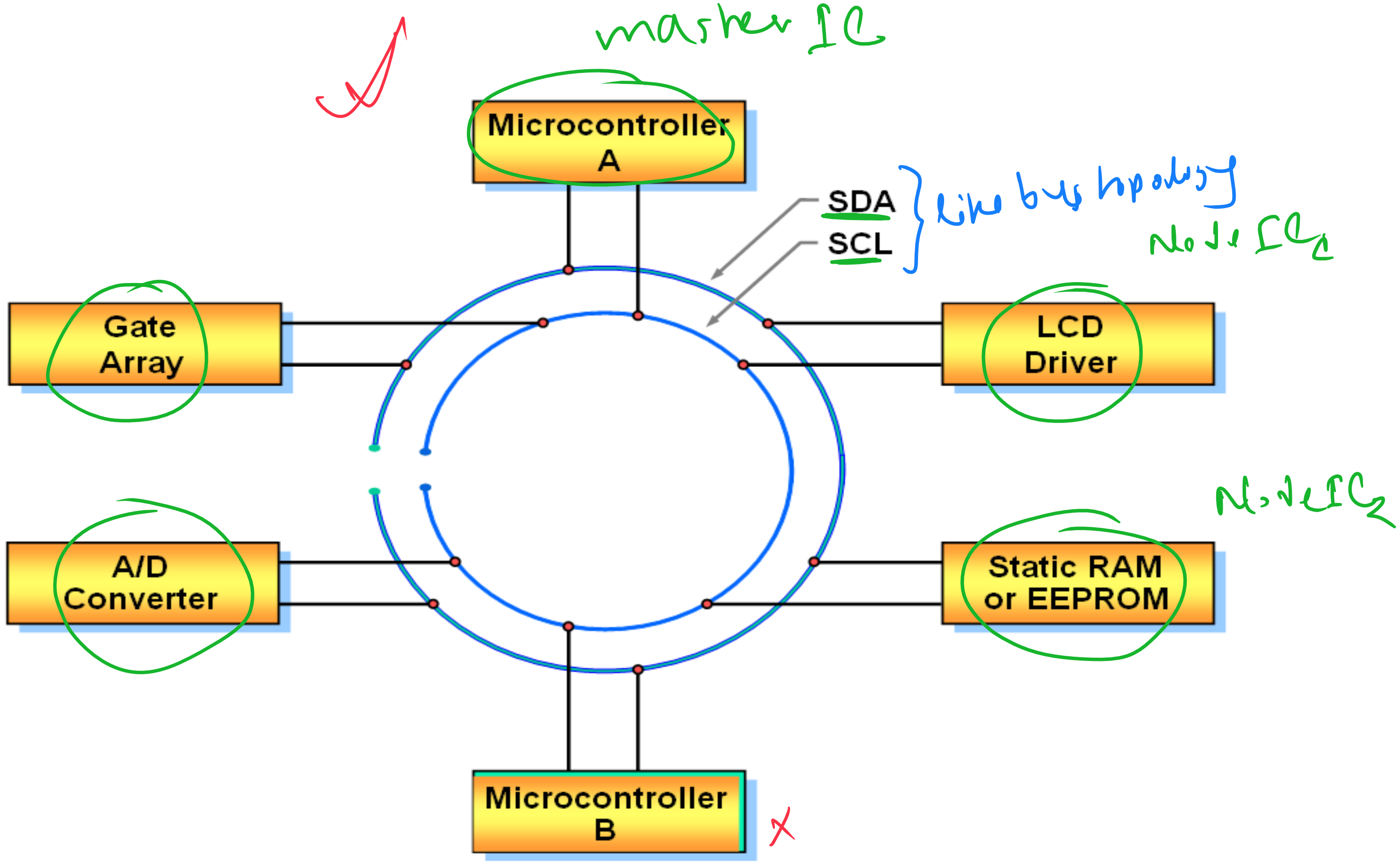
- **True multi-master capability**
  - Clock synchronization
  - Arbitration procedure → who will act as master
- **Transmission speeds up to 100Khz (classic I^2C)**
- **Allows series resistor for IC protection**
- **Compatible with different IC technologies**

as IC may vary from manufacturer to manufacturer

I²C Bus Definitions

- **Master:**
  - Initiates a transfer by generating **start** and **stop** conditions
  - Generates the clock pulses
  - Transmits the slave address
  - Determines data transfer direction
- **Slave:**
  - Responds only when addressed by master → master determines who it should communicate with
  - Timing is controlled by the clock line → scl line

I^{2}C Bus Configuration Example

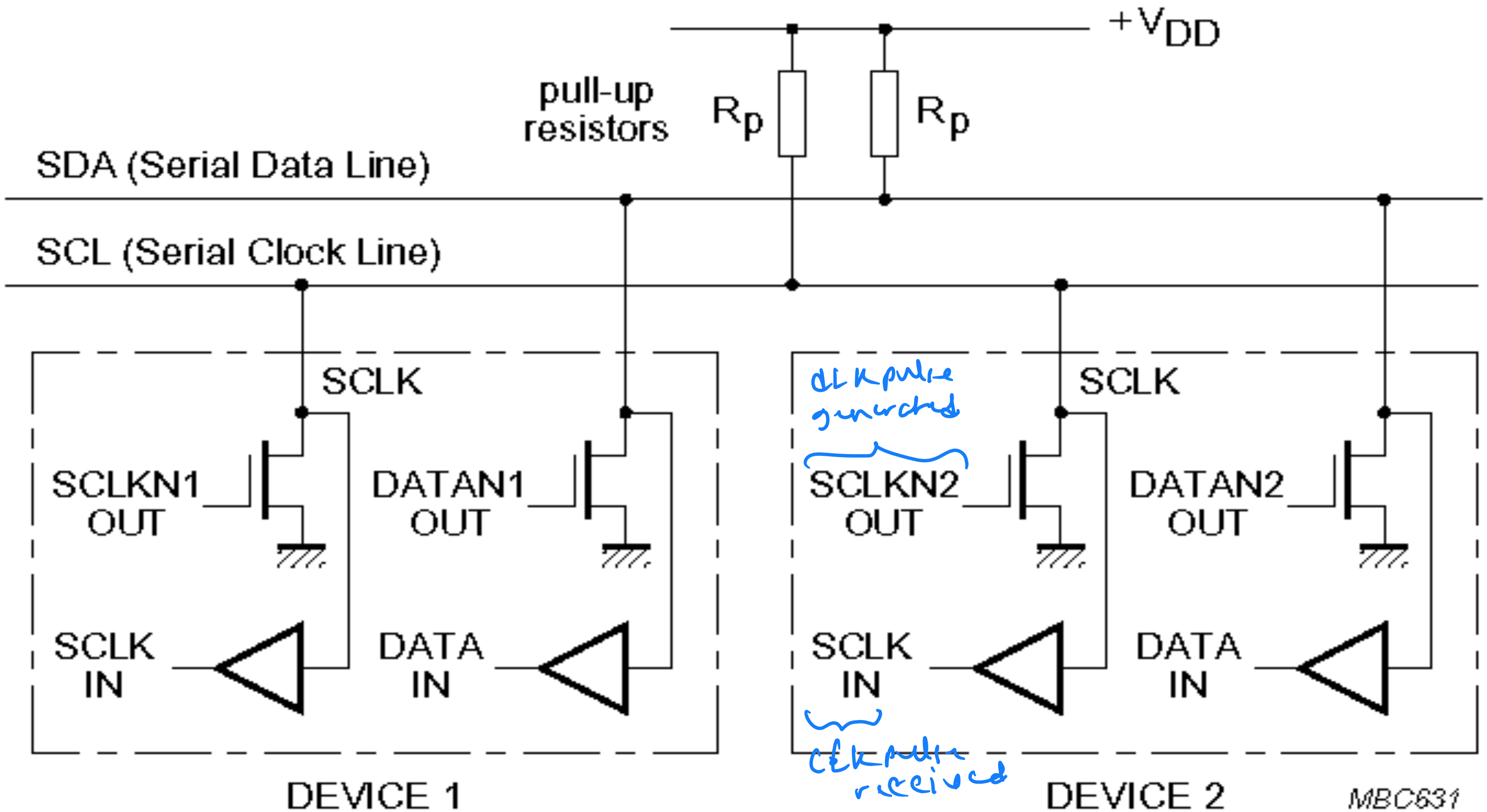


I^2C Hardware Details

- **Devices connected to the bus must have an open drain or open collector output for serial clock and data signal**
- **The device must also be able to sense the logic level on these pins**
- **All devices have a common ground reference**
- **The serial clock and data lines are connected to Vdd(typically +5V) through pull up resistors**
- **At any given moment the I^2C bus is:**
  - Quiescent (Idle), or
  - in Master transmit mode or
  - in Master receive mode.

either one of these three modes

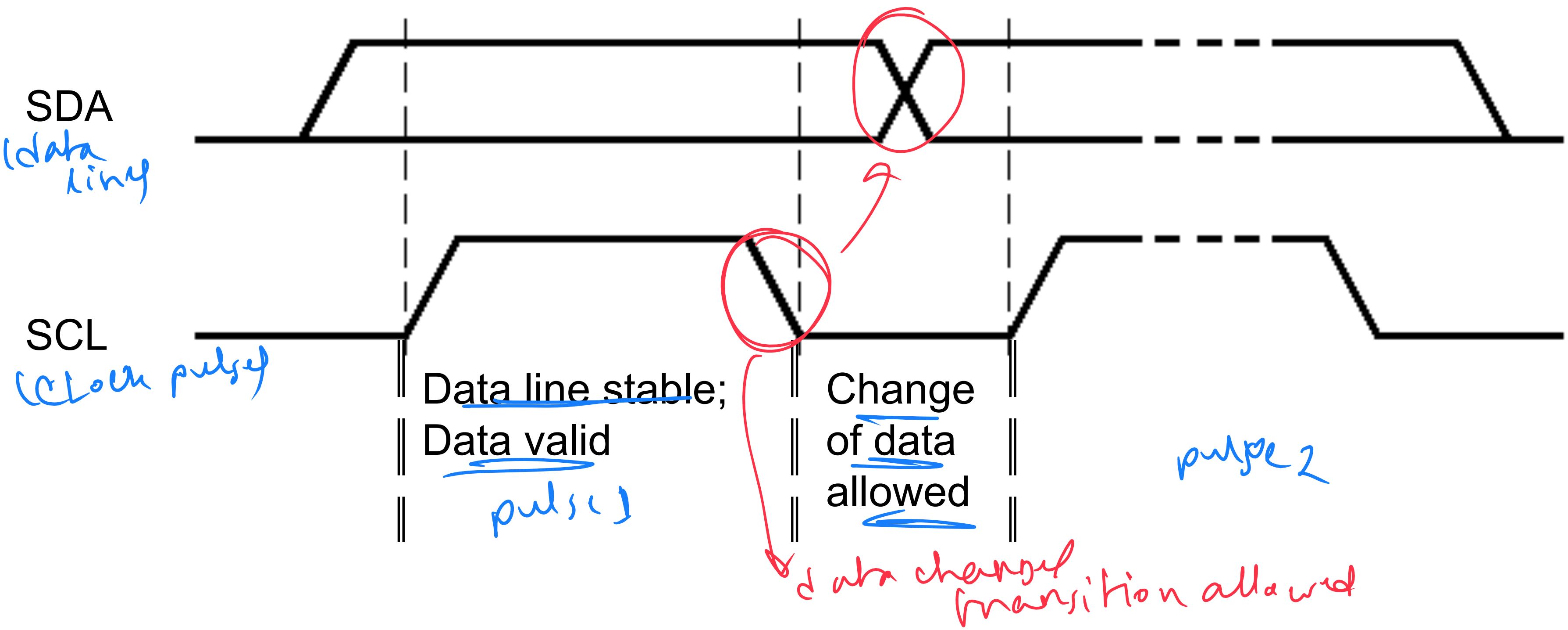
I^2C Electrical Aspects



- I^2C devices are wire ANDed together.
- If any single node writes a zero, the entire line is zero

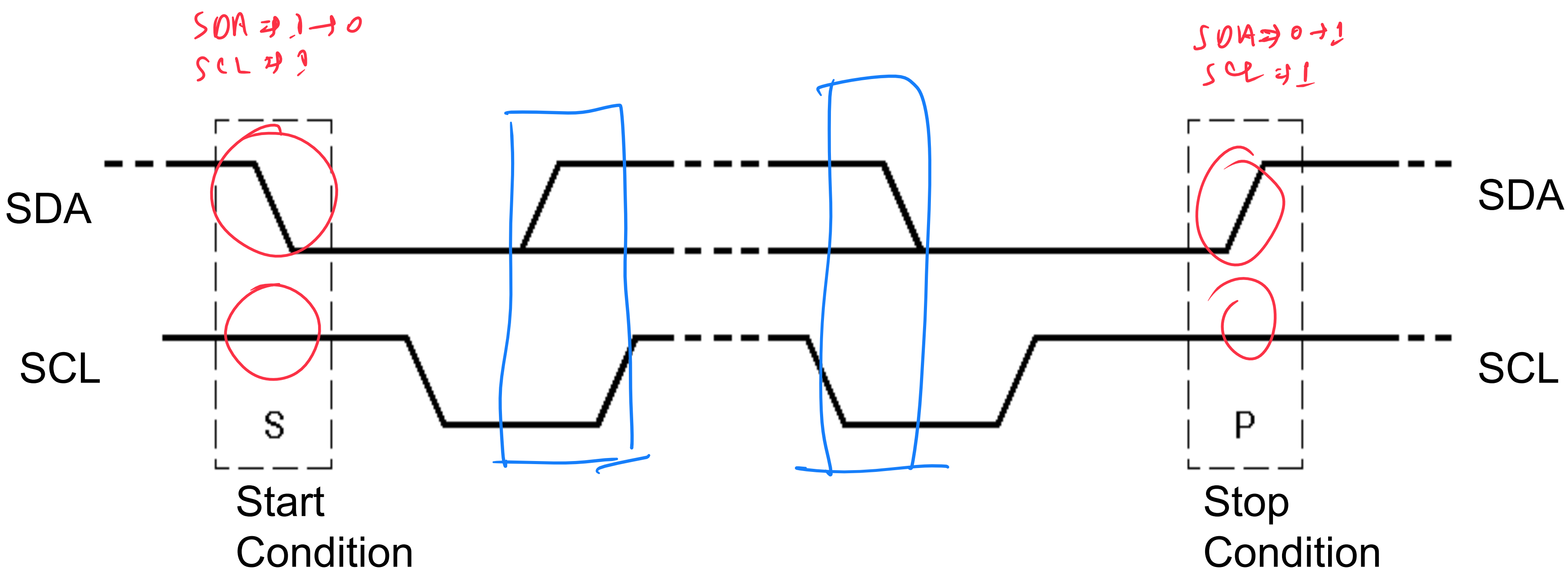
Bit Transfer on the I^{2}C Bus

- In normal data transfer, the data line only changes state when the clock is low



Start and Stop Conditions

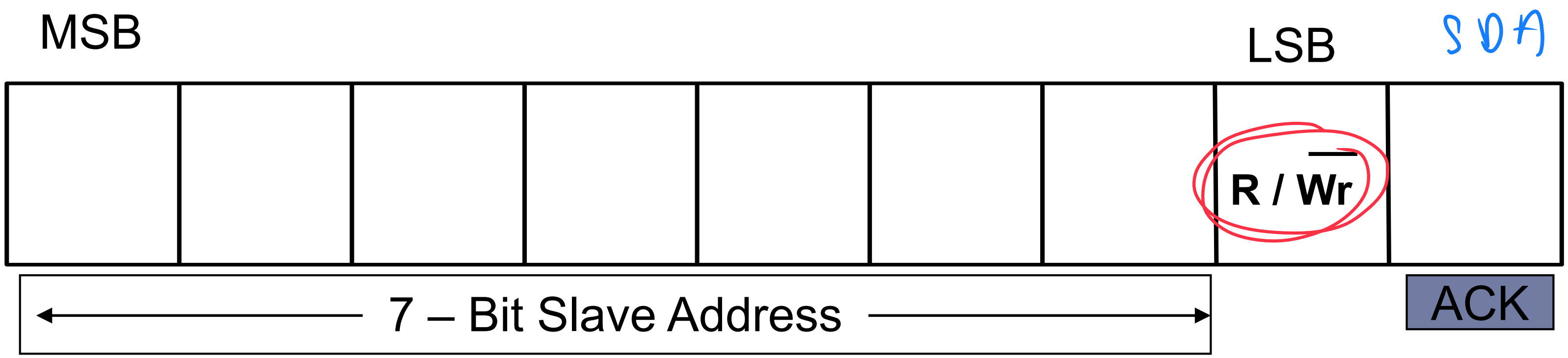
- A transition of the data line while the clock line is **high** is defined as either a start or a stop condition.
- Both **start** and **stop** conditions are generated by the bus master
- The bus is considered busy after a **start** condition, until a **stop** condition occurs



I^2C Addressing

- **Each node has a unique 7 (or 10) bit address**
- **Peripherals often have fixed and programmable address portions**
- **Addresses starting with 0000 or 1111 have special functions:-**
  - 0000000 Is a General Call Address
  - 0000001 Is a Null (CBUS) Address
  - 1111XXX Address Extension
  - 1111111 Address Extension – Next Bytes are the Actual Address

First Byte in Data Transfer on the I^{2}C Bus



R/Wr

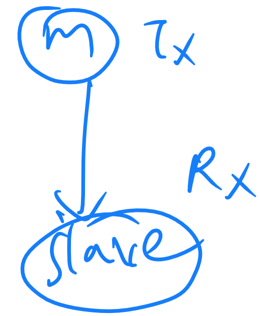
- 0 – Slave written to by Master → master will transmit
- 1 – Slave read by Master → master will receive

ACK – Generated by the slave whose address has been output.

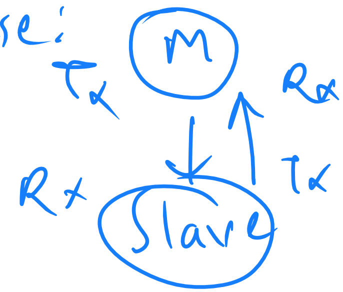
I^{2}C Bus Connections

- **Masters can be**
  - Transmitter only
  - Transmitter and receiver
- **Slaves can be**
  - Receiver only
  - Receiver and transmitter

1st case:

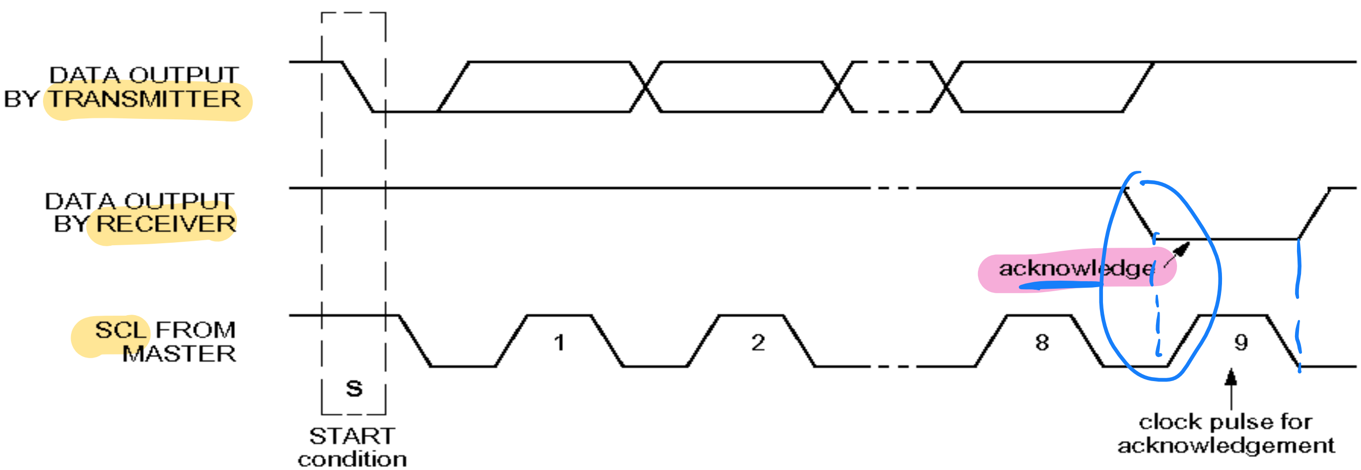


2nd case:



Acknowledgements

- Master/slave receivers pull data line low for one clock pulse after reception of a byte
- Master receiver leaves data line high after receipt of the last byte requested
- Slave receiver leaves data line high on the byte following the last byte it can accept



Transmitter releases SDA line during 9th clock pulse.

Acknowledgement from receiver

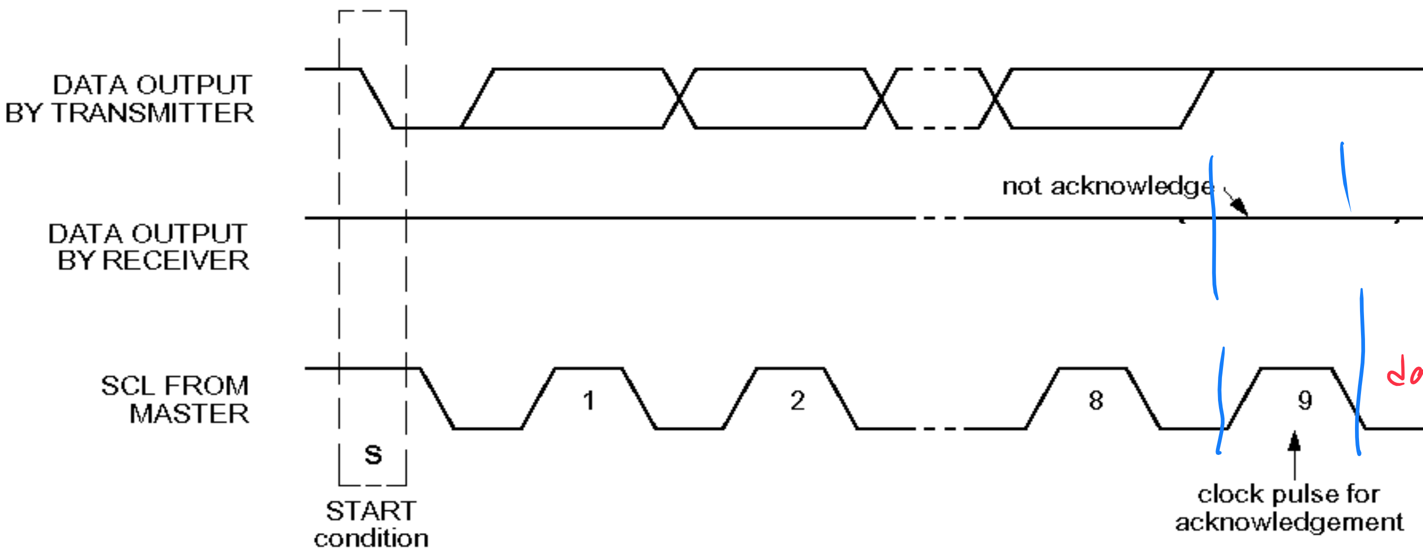
data line low for ninth bit

Acknowledgements → when they can be sent

- **From Slave to Master Transmitter:**
  - After address received correctly
  - After data byte received correctly
- **From Slave to Master Receiver:**
  - Never (Master Receiver generates ACK)
- **From Master Transmitter to Slave:**
  - Never (Slave generates ACK)
- **From Master Receiver to Slave:** (Master Receiver, receives data byte from slave, and sends ACK)
  - After data byte received correctly

Negative Acknowledge

- Receiver leaves data line high for one clock pulse after reception of a byte



Transmitter releases SDA line during 9^{th} clock pulse.

Not acknowledgement (NACK) from receiver

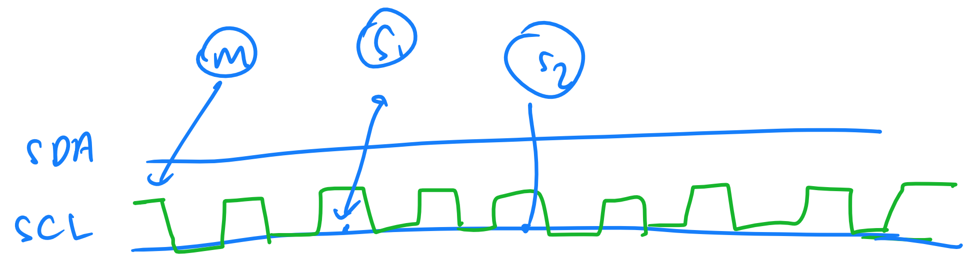
data line high for NACK on the 9th bit

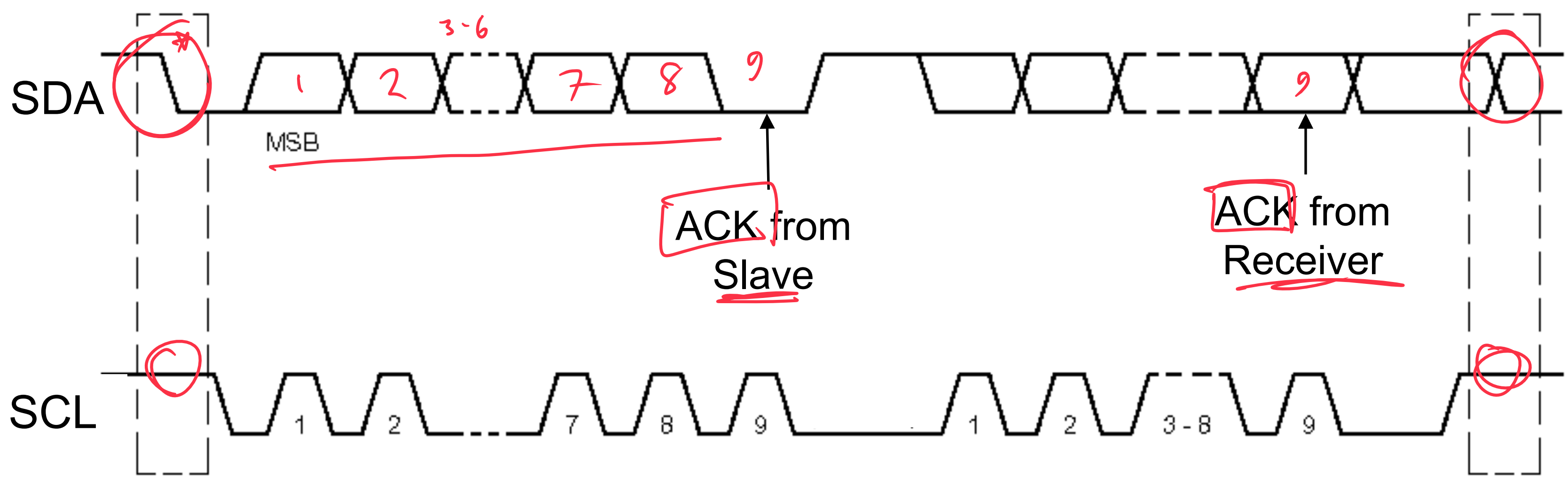
Negative Acknowledge (Cont'd.) (same as Ack)

- **From Slave to Master Transmitter:**
  - After address not received correctly
  - After data byte not received correctly
  - Slave Is not connected to the bus
- **From Slave to Master Receiver:**
  - Never (Master Receiver generates ACK)
- **From Master Transmitter to Slave:**
  - Never (Slave generates ACK)
- **From Master Receiver to Slave:**
  - After last data byte not received correctly

Data Transfer on the I²C Bus

- **Start Condition**
- **Slave address + R/W**
  - Slave acknowledges with ACK
- **All data bytes**
  - Each followed by ACK
- **Stop Condition**





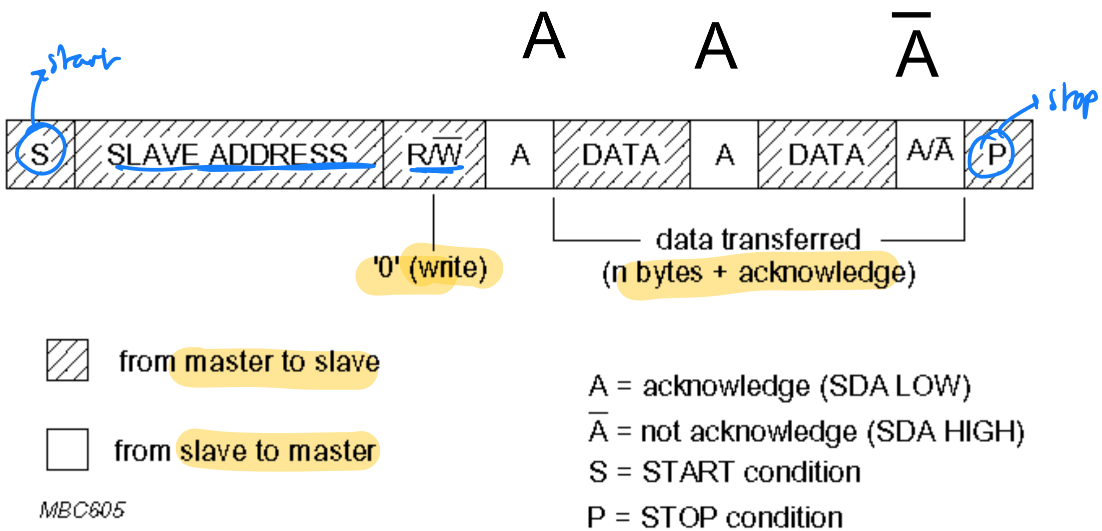
Start

Remember : Clock is produced by Master

Stop

Data Formats

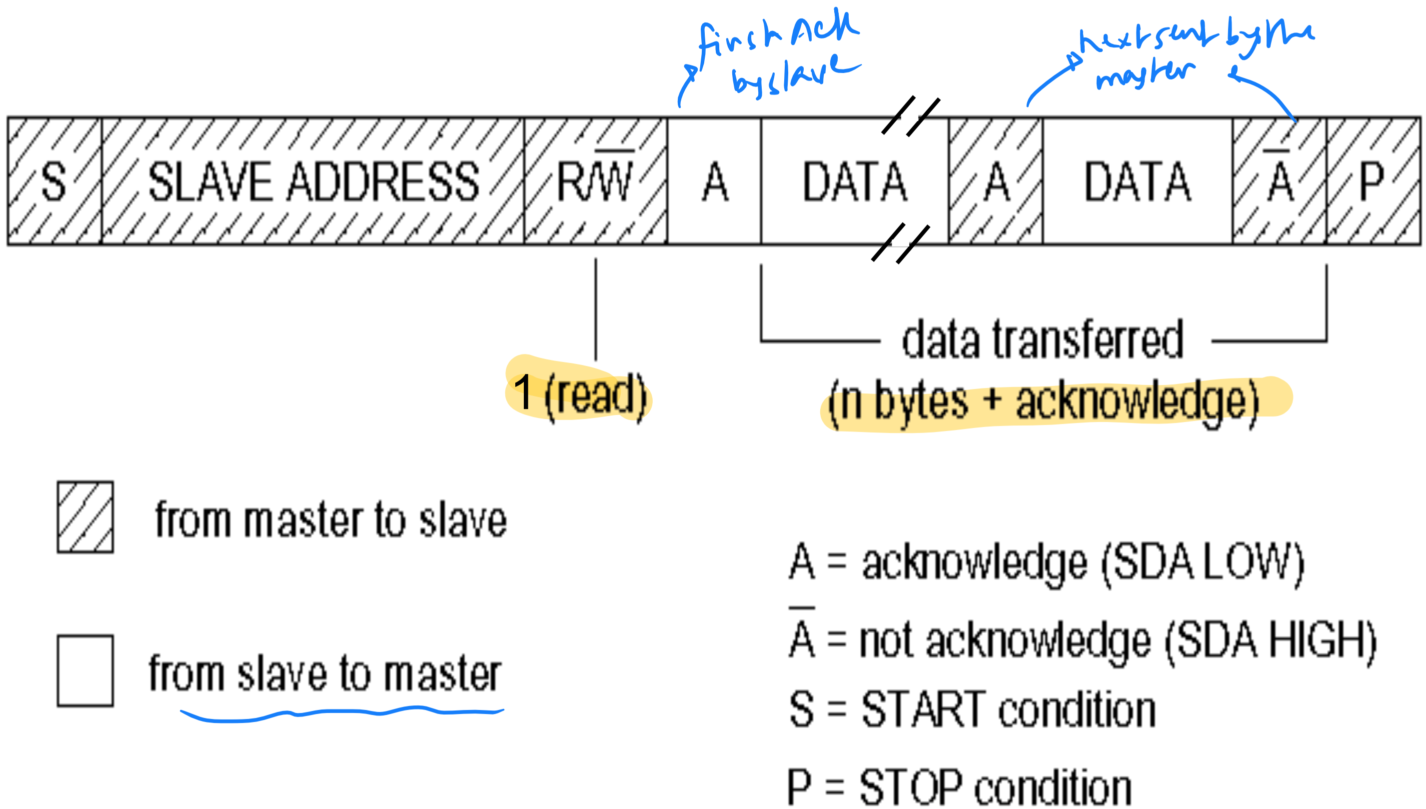
- **Master writing to a Slave**



Data Formats Cont'd.

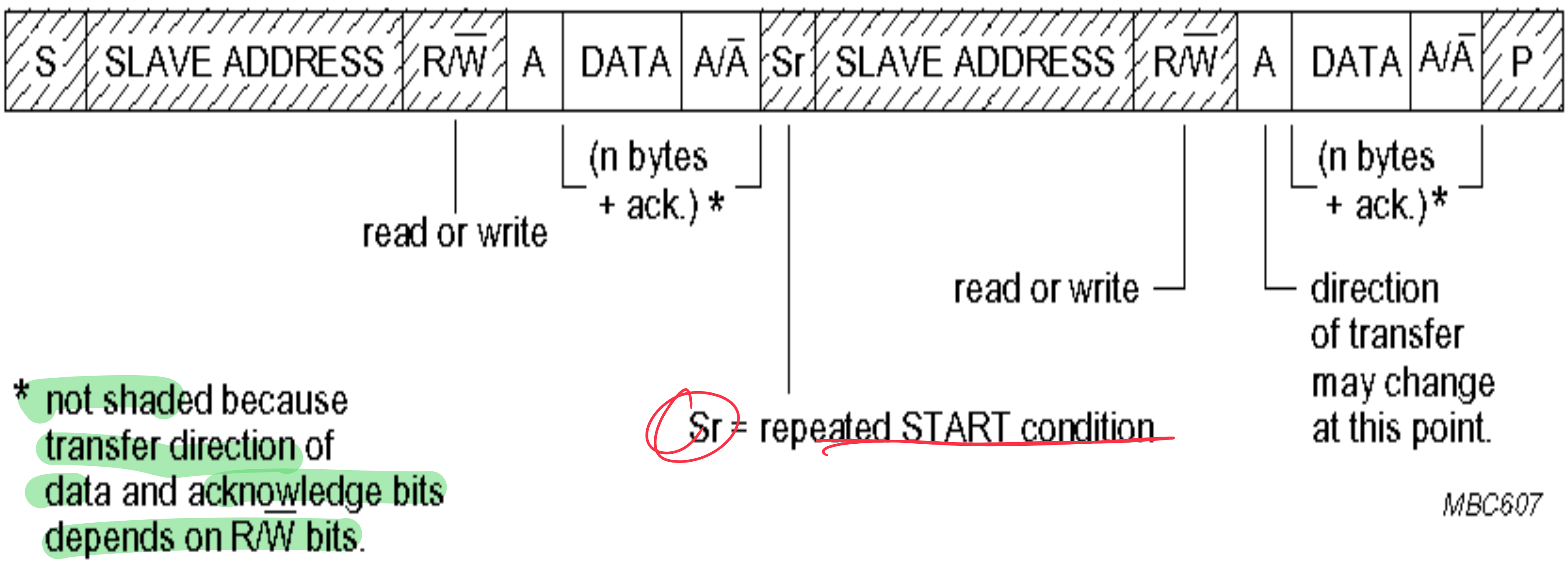
- **Master reading from a Slave** :

Master is Receiver of data and Slave is Transmitter of data.



Data Formats Cont'd.

Combined Format

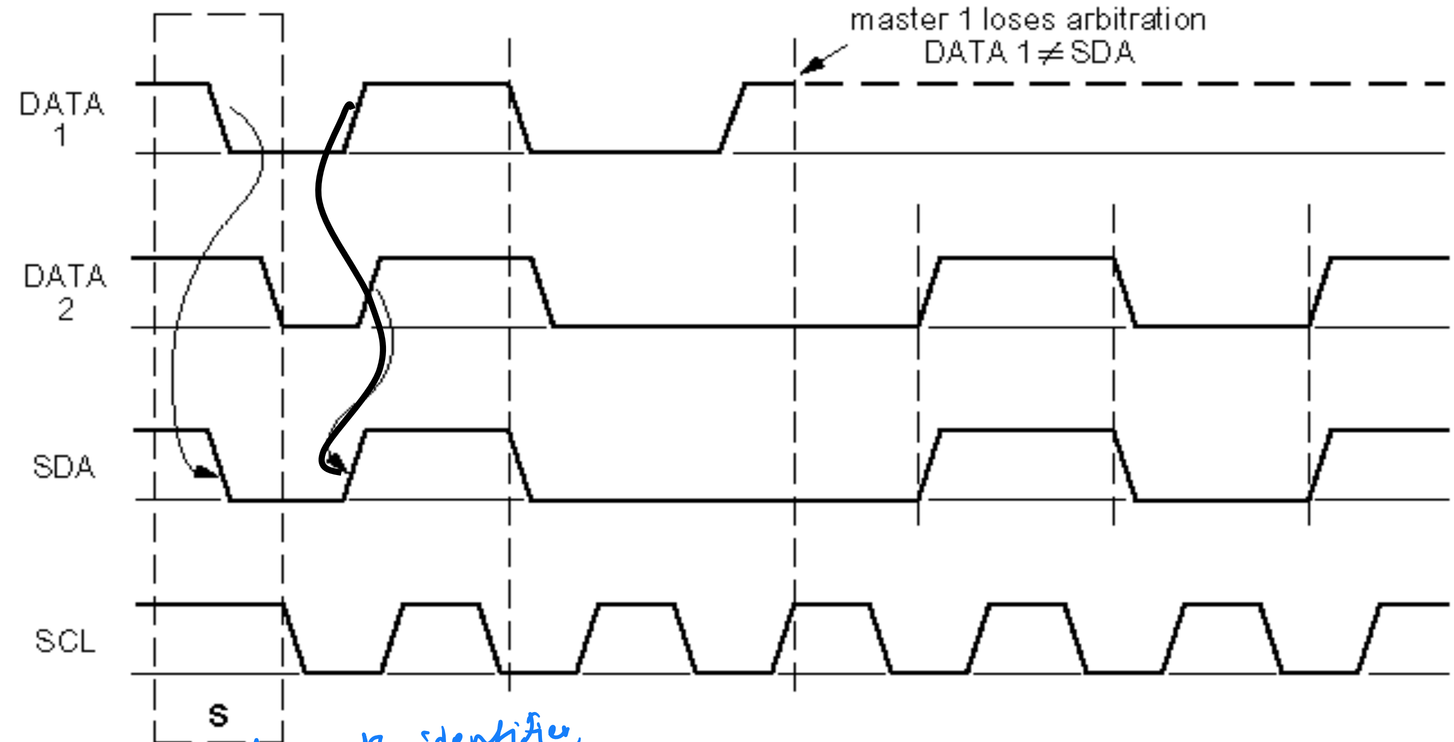


- A **repeated start** avoids releasing the bus and therefore prevents another master from taking over the bus

Multi-master I^{2}C Systems

- **Multimaster situations require two additional features of the I^{2}C protocol**
- **Arbitration:**
  - Arbitration is the procedure by which competing masters decide final control of the bus
  - I^{2}C arbitration does not corrupt the data transmitted by the prevailing master
  - Arbitration is performed bit by bit until it is uniquely resolved
  - Arbitration is lost by a master when it attempts to assert a **high** on the data line and fails

Arbitration Between Two Masters



depending on the identifier, one loses and one gains the arbitration

- As the data line is like a wired-AND, a ZERO address bit overwrites a ONE
- The node detecting that it has been overwritten stops transmitting and waits for the Stop Condition before it retries to arbitrate the bus

Error Checking

- **I²C defines the basic protocol and timing**
  - Protocol errors are typically flagged by the interface
  - Timing errors may be flagged, or in some cases could be interpreted as a different bus event
- **Microprocessors communicating with each other can add a checksum or equivalent**

Bus Recovery

- **An I²C bus can be "locked" when:**
  - A Master and a Slave get out of synch
  - A Stop is omitted or missed (possibly due to noise)
  - Any device on the bus holds one of the lines low improperly, for any reason
  - A shorted bus line
- **If SCL can be driven, the Master may send extra clocks until SDA goes high, then send a Stop.** If no stop condition/stuck in one condition
- **If SCL is stuck low, only the device driving it can correct the problem.**

Available I²C Devices

- **Analog to Digital Converters (A/D, D/A):** MMI functions, battery & converters, temperature monitoring, control systems
- **Bus Controller:** Telecom, consumer electronics, automotive, Hi-Fi systems, PCs, servers
- **Bus Repeater, Hub & Expander:** Telecom, consumer electronics, automotive, Hi-Fi systems, PCs, servers
- **Real Time Clock (RTC)/Calendar:** Telecom, EDP, consumer electronics, clocks, automotive, Hi-Fi systems, FAX, PCs, terminals
- **DIP Switch:** Telecom, automotive, servers, battery & converters, control systems
- **LCD/LED Display Drivers:** Telecom, automotive instrument driver clusters, metering systems, POS terminals, portable items, consumer electronics

Available I^{2}C Devices

- **General Purpose Input/Output (GPIO) Expanders and LED Display Control:** Servers, keyboard interface, expanders, mouse track balls, remote transducers, LED drive, interrupt output, drive relays, switch input
- **Multiplexer & Switch:** Telecom, automotive instrument driver clusters, metering systems, POS terminals, portable items, consumer electronics
- **Serial RAM/ EEPROM:** Scratch pad/ parameter storage
- **Temperature & Voltage Monitor:** Telecom, metering systems, portable items, PC, servers
- **Voltage Level Translator:** Telecom, servers, PC, portable items, consumer electronics

End use (all small case)

- **Telecom:** Mobile phones, Base stations, Switching, Routers
- **Data processing:** Laptop, Desktop, Workstation, Server
- **Instrumentation:** Portable instrumentation, Metering systems
- **Automotive:** Dashboard, Infotainment
- **Consumer:** Audio/video systems, Consumer electronics (DVD, TV etc.)

I^2C designer benefits

- Functional blocks on the block diagram correspond with the actual ICs; designs proceed rapidly from block diagram to final schematic
- No need to design bus interfaces because the I^2C-bus interface is already integrated on-chip
- Integrated addressing and data-transfer protocol allow systems to be completely software-defined
- The same IC types can often be used in many different applications

I2C designer benefits

- Design-time improves as designers quickly become familiar with the frequently used functional blocks represented by I^2C-bus compatible ICs
- ICs can be added to or removed from a system without affecting any other circuits on the bus
- Fault diagnosis and debugging are simple; malfunctions can be immediately traced
- Software development time can be reduced by assembling a library of reusable software modules
- The simple 2-wire serial I^2C-bus minimizes interconnections so ICs have fewer pins and there are fewer PCB tracks; resulting in smaller and less expensive PCBs

I^2C Manufacturers benefits

- The completely integrated I^2C-bus protocol eliminates the need for address decoders and other 'glue logic'
- The multi-master capability of the I^2C-bus allows rapid testing/alignment of end-user equipment via external connections to an assembly-line
- Increases system design flexibility by allowing simple construction of equipment variants and easy upgrading to keep design up-to-date
- The I^2C-bus is a de facto world standard that is implemented in over 1000 different ICs (Philips has > 400) and licensed to more than 70 companies

Thank You !!